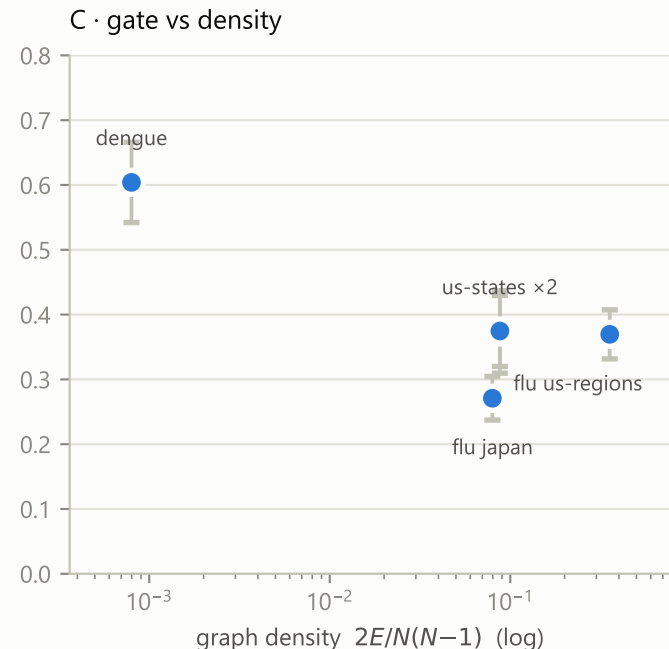
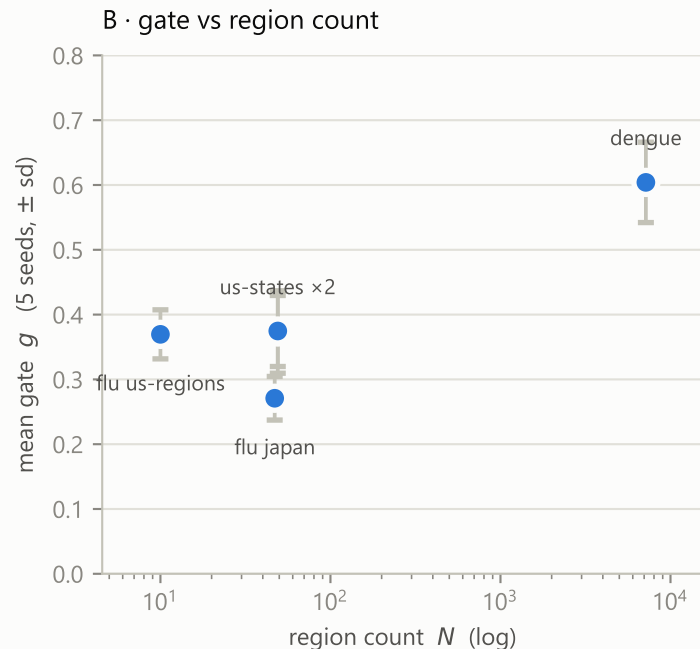
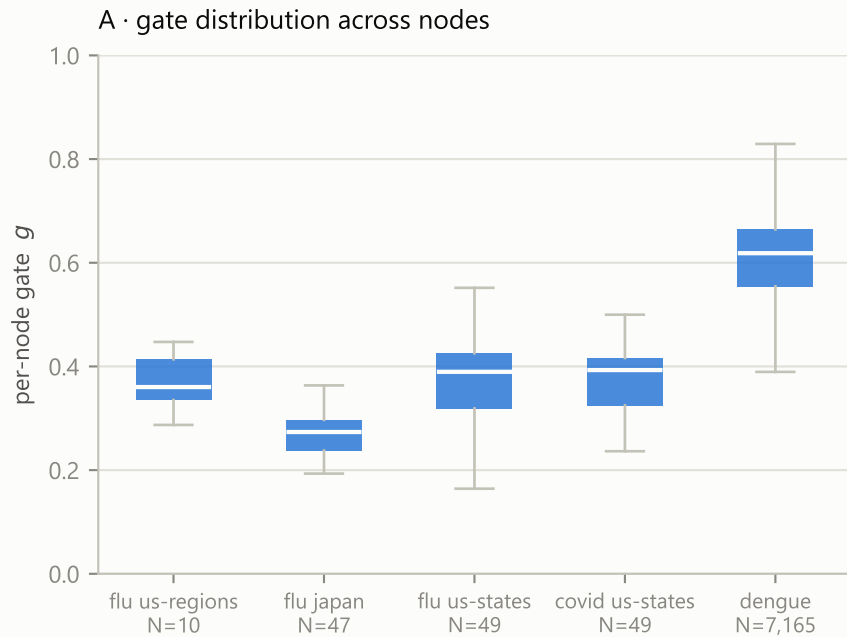




Learned spatial gate g , single-disease encoder, 5 seeds per panel. Error bars are the seed sd; box A is the per-node spread pooled over seeds (IQR, whiskers $1.5 \times \text{IQR}$, fliers hidden).

flu us-states and covid us-states share an identical graph ($N=49$, 206 edges) and land on the same gate (0.373 vs 0.375) — the gate reads graph structure, not disease. Mean degree is 3.2–5.7 across all five panels, so density $\approx 1/N$ here: B and C are one ordering shown twice, not independent evidence. The four panels below $N=50$ are within noise of each other except japan; the rise is carried by dengue alone, so B and C show a range across graph sizes, not a fitted trend over five points.